

Rachit Sabharwal

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Professional Summary

Six years building production data pipelines, predictive models, and stakeholder-facing data products on real-world healthcare data — EHR, claims, and multi-site clinical records. Recent work centers on LLM-based automation of free-text classification and document workflows, alongside SQL/Python ETL, CI/CD-automated reporting, and R Shiny / Tableau decision-support tools. Comfortable owning work end-to-end and partnering with operational and clinical leadership on roadmap-shaping decisions.

Education

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| Ph.D. The University of Texas Health Science Center at Houston, Epidemiology and Biostatistics
• Certificates: Advanced Data Science | Houston, TX
Aug 2022 – Aug 2026 |
| Cert. The University of California, Berkeley, Software Development and Programming | Berkeley, CA
June 2023 – Apr 2025 |
| M.S. The University of Texas Health Science Center at Houston, Biostatistics | Houston, TX
Jan 2020 – May 2022 |
| B.S. University of Rochester, Environmental Health | Rochester, NY
Sept 2014 – May 2018 |

Experience

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| The University of Texas Health Science Center at Houston , Research Data Scientist, School of Public Health
<ul style="list-style-type: none">Contributed claims-data and EHR analyses to an R01 grant proposal on disability in AYA cancer and birth-defect survivors; applied regression, survival analysis, and linear mixed models to population-based cohorts and co-authored 4 peer-reviewed publications on vaccine non-response and pediatric COVID comorbidities.Led a team evaluating LLM feasibility for free-text physician-note classification in the multi-site DETECT-RPC study, shaping the project's NLP pipeline design; built an agentic LLM workflow that migrated the r4epi epidemiology textbook from R Markdown to Quarto, chaining file transfer, code reformatting, and grammar transformation across hundreds of files.Owned end-to-end recurring reporting for TexasCARES — redesigned the SQL ETL to cut data-munging turnaround 24x, then stood up CI/CD-automated pipelines that ingest new data weekly, run quality checks on PII-handled cohorts, and publish stakeholder outputs without manual intervention; backed by a full lifecycle test suite.Designed the multi-site star schema and SQL ETL for DETECT-RPC, harmonizing clinical records from 6 academic medical centers; shipped a public statewide COVID-19 dashboard (Python, Tableau) and an R Shiny CWDA→ICD-10 mapping tool adopted for cross-team use, plus a Quarto dashboard giving PIs real-time visibility into multi-site data collection.Supervised 2 M.S. students on quantitative analyses and 2 more on automated-report QA/QC; led a 2-person TA team for PH 1976 (Fundamentals of Data Analytics and Predictions). | Houston, TX
Feb 2020 – present |
| The Dow Chemical Company , ML R&D Intern, MiLOS, Core R&D
<ul style="list-style-type: none">Shipped an R Shiny decision-support app that automated 50% of enterprise Life Cycle Assessment workflows, estimated at \$15M in annual cost savings across global operations.Benchmarked predictive models across 25+ measurement-error scenarios using calibration metrics to quantify forecast reliability for non-technical stakeholders; authored an internal white paper adopted as organizational guidance. | Lake Jackson, TX
May 2025 – Aug 2025 |
| Genentech , Informatics R&D Intern
<ul style="list-style-type: none">Benchmarked a published multimodal GNN on DILI prediction and prototyped a simpler MLP alternative, quantifying the complexity-vs-performance tradeoff for R&D stakeholders. | San Francisco, CA
May 2021 – Jan 2022 |

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| Procter & Gamble , Advanced Analytics Co-Op, Consumer & Market Knowledge | Cincinnati, OH |
| <ul style="list-style-type: none"> Delivered predictive and explanatory regression models on shopper and retailer behavior for global Fabric Care brand teams, informing a corporate strategy shift toward DTC ecommerce. Raised cross-functional analyst throughput 1.5x by building GCP big-data ETL pipelines on Apache Airflow. | Jan 2021 – May 2021 |
| Bristol Myers Squibb , Data Engineering Intern | San Francisco, CA |
| <ul style="list-style-type: none"> Built PatRec, a full-stack recommendations app chaining chemical-entity NER, federated search, and TF-IDF similarity ranking to surface relevant patents from an 80M-patent corpus for internal scientists. Designed relational (PostgreSQL) and graph (Neo4j) database schemas supporting multiple internal R&D analytical workloads. | June 2020 – Aug 2020 |

Selected Publications

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| Trust and Uncertainty Quantification in Machine Learning Models Under Measurement Error
Sabharwal R
(The Dow Chemical Company, Internal White Paper) | Aug 2025 |
| Baseline characteristics of SARS-CoV-2 vaccine non-responders in a large population-based sample
Yaseen A, DeSantis SM, Sabharwal R , et al.
10.1371/journal.pone.0303420 (PLoS One) | May 2024 |
| Scholarly recommendation systems: a literature survey
Zhang Z, Patra BG, Sabharwal R , et al.
10.1007/s10115-023-01901-x (Knowledge and Information Systems) | June 2023 |

Skills

Programming & Data: SQL, Python (polars, pandas, duckdb), R (data.table, tidyverse, tidymodels), Bash

AI & LLM Systems: Agentic LLM pipelines, LLM APIs (Groq, Claude, Gemini, OpenAI-compatible), free-text classification, document tokenization, NLP, recommender systems

Statistics & ML: Regression (linear, logistic, GLM), Survival Analysis, Linear Mixed Models, A/B Testing, Causal Inference, Bayesian Inference & UQ, Measurement Error Modeling, scikit-learn, PyTorch, Hugging Face

Data Platforms & Pipelines: PostgreSQL, BigQuery, Apache Airflow, GitHub/GitLab CI/CD, version-controlled SQL pipelines, Docker, Kubernetes, GCP, AWS, Azure, REDCap

Visualization & Reporting: Tableau, R Shiny, Quarto, Streamlit, Python Dash, Plotly, ggplot2, DT

Domain: Real-World Data (EHR, claims, surveys), Healthcare Operations, Multi-Site Data Harmonization, Public Health Surveillance, Consumer Analytics, Regulated/PII Data Handling